

MEMO

From: Mark McIntyre, PharmD, Lead PharmAchieve MCQ Instructor:

To: MCQ Class

In class, some of you have asked me which inhibitors and inducers: what they are, and which ones to remember.

Consider phase I enzymes, CYP450 being the most common example, as a mechanism by which exogenous substances such as medications are made easier to excrete by the body. The medications metabolized by these enzymes are called substrates.

Most subcategories of these phase I enzymes have compounds which may increase or decrease their efficiency in catalyzing drug metabolism. Medications that reduce the capacity for the body to metabolize a substance are called inhibitors whereas medications that increase metabolism are called inducers.

Example 1: A patient with hypercholesterolemia is stabilized on atorvastatin 40mg PO daily, a CYP450 3A4 substrate. The patient then develops an active infection with tuberculosis requiring combination antimicrobial therapy including rifampin. Rifampin is a very potent enzyme inducer of CYP450 3A4. The consequence of this enzyme induction would be an increased metabolism of atorvastatin and a resultant reduction in atorvastatin concentrations. This may lead to lack of efficacy and increased cholesterol and/or vascular events.

As several hundred enzyme substrates, inducers and inhibitors exist, there can be no expectation in the exam that you should memorize all of them. However, there are several that are either common or severe that are worth committing to memory. The list attached should help out by highlighting the medications most commonly seen in clinical practice.

Lastly, there a number of other enzymes that may act as transporter "pumps" such as p-GP, OAT and various others. These transporter proteins have specific substrates and may be inhibited by select medications. They are rarely induced. The clinical consequence of inhibition remains largely the same as CYP interactions but for a different reason. CYP enzymes generally metabolize medications. P-GP and OAT transporter proteins will move medications from the intestine into the circulation. I have included a short list in the attached document as well for your information.

	3A4	2D6	2C9	2C19	1A2	p-GP
Substrate	Sildenafil Midazolam Diazepam Atorvastatin Simvastatin Felodipine Fentanyl Cyclospoinc	Dextromethorphan Metoprolol Venlafaxine	Celecoxib Warfarin Phenytoin	Phenytoin	Theophylline Caffeine Duloxetine	Digoxin Colchicine Dabigatran
inhibitor	Clarithromycin Itraconazole Voriconazole Ritonavir Grapefruit juice	Bupropion Fluoxetine Paroxetine	Amiodarone Fluconazole	Fluvoxamine Fluconazole Omeprazole	Fluvoxamine Ciprofloxacin	Ritonavir Cyclosporine Amiodarone Dronedarone
Inducer	Carbamazepine Phenytoin Phenobarbital Rifampin St. John's Wort	None	Rifampin Carbamazepine	Rifampin	Phenytoin Smoking	Rifampin St. John's Wort

Licensed to Shreyansh Salunke (ID: 1477)